CHAPTER 12

HYDROSTATICS

The study of fluids at rest.

DENSITY

Density (ρ) is a measure of the mass per unit volume (kg/m^3).

$$t/m^3 = kg/l = g/ml = g/cm^3$$

The relative density of a substance is the ratio of the mass of that substance to the mass of an equal volume of pure water, in other words, it is the ratio of the density of the substance to that of pure water.

Densities of liquids may be measured by a hydrometer. This is an instrument consisting of a graduated stem with a hollow bulb float at the lower end of the stem, and keel beneath the bulb so that the hydrometer floats upright when placed in a liquid. The stem is graduated, usually in g/ml, and this figure is read off at the level of the liquid at the stem.

MIXING OF LIQUIDS OF DIFFERENT DENSITIES

When liquids of different densities are mixed together it is usual to assume that their volumes and masses are not affected due to mixing, that is, the final volume of the whole mixture is equal to the sum of the volumes of each constituent before mixing, and the final mass of the whole mixture is equal to the sum of the masses of each constituent before mixing.

Example. 2 g of oil of relative density 0·8 are mixed with 4 g of oil of relative density 0·9. Find the relative density of the mixture.

$$\text{Total mass of mixture} = 2 + 4 = 6 \text{ g} \quad \ldots \quad \ldots \quad \ldots (i)$$

As the densities of the oils are 0·8 and 0·9 g/ml respectively, and volume = mass ÷ density, then,

$$\text{Total volume} = \frac{2}{0{\cdot}8} + \frac{4}{0{\cdot}9} \text{ ml}$$

$$= \frac{1{\cdot}8 + 3{\cdot}2}{0{\cdot}72}$$

$$= \frac{5}{0{\cdot}72} \text{ ml} \quad \dots \quad \dots \quad \dots (ii)$$

$$\text{Density of mixture} = \frac{\text{Total mass}}{\text{Total volume}}$$

$$= \frac{6 \times 0{\cdot}72}{5}$$

$$= 0{\cdot}864 \text{ g/ml}$$

$$\text{Relative density} = 0{\cdot}864 \text{ Ans.}$$

Example. Three liquids of relative density 0·75, 0·85 and 0·95 are mixed in the proportion of 2, 3 and 4, by volume. Find the relative density of the mixture.

As the proportion is by volume, let the volumes be 2, 3 and 4 ml respectively.

Densities of the oils are 0·75, 0·85 and 0·95 g/ml respectively.

$$m = V \times \rho$$

$$\text{Total mass} = 2 \times 0{\cdot}75 + 3 \times 0{\cdot}85 + 4 \times 0{\cdot}95$$

$$= 1{\cdot}5 + 2{\cdot}55 + 3{\cdot}8$$

$$= 7{\cdot}85 \text{ g}$$

$$\text{Total volume} = 2 + 3 + 4$$

$$= 9 \text{ ml}$$

$$\text{Density} = \frac{\text{Total mass}}{\text{Total volume}}$$

$$= \frac{7{\cdot}85}{9}$$

$$= 0{\cdot}872 \text{ g/ml}$$

$$\text{Relative density} = 0{\cdot}872 \text{ Ans.}$$

APPARENT LOSS OF WEIGHT OF A SUBMERGED BODY

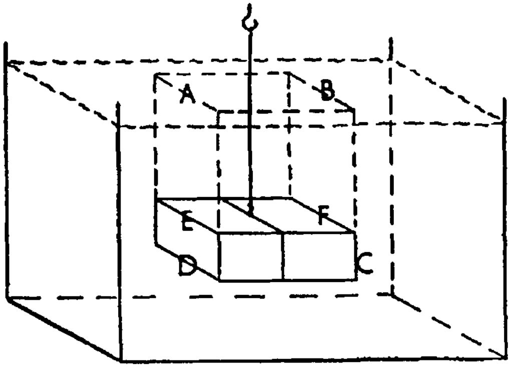

Fig. 203

Consider a vertical column ABCD of the liquid in a tank as shown in Fig. 203 (before a body is submerged). The column of liquid is in equilibrium therefore the upward supporting thrust on its base DC is equal to the downward weight of the column.

When a body is lowered and submerged in the liquid such as EFCD, the upward thrust at the same level DC (that is on the base of the body), must be the same as before but there is also a downward force on the top of the body due to the column of liquid ABFE above it. Therefore the net force on the body is the difference between the upward supporting force on the base and the downward weight of liquid on the top, this is equal to the difference between the columns of liquid ABCD and ABFE which is the weight of the liquid EFCD displaced by the body.

Hence, the upward force of buoyancy on a submerged body is equal to the weight of liquid displaced by the body. This is simply verified experimentally by suspending a regular shaped body (whose volume can be calculated) from a spring balance, noting the weight when the body is in air and again when lowered into water until it is submerged, the difference in weights read from the spring balance is the 'apparent loss of weight' which will be found to be equal to the weight of water displaced (that is the weight of water of equal volume to the body).

This principle also provides a useful method of determining the relative density of a solid heavier than water as shown by the following.

Example. A piece of cast iron registers a weight of 56 N in air and 48·23 N when suspended in fresh water. Find the relative density of this cast iron.

$$\text{Weight of water displaced} = 56 - 48{\cdot}23$$
$$= 7{\cdot}77 \text{ N}$$

The volume of water displaced is the same as the volume of the cast iron, therefore,

$$\text{Relative density} = \frac{\text{Wt. of cast iron}}{\text{Wt. of an equal volume of fresh water}}$$
$$= \frac{56}{7{\cdot}77}$$
$$= 7{\cdot}207 \text{ Ans.}$$

From the above it can be seen that:

$$\text{Relative density} = \frac{\text{Wt. in air}}{\text{Wt. in air} - \text{wt. in fresh water}}$$

Example. A block of aluminium of 11·5 kg mass is suspended from a wire 1·5 mm diameter and lowered until submerged into a tank containing oil of relative density 0·9. Taking the relative density of aluminium as 2·56, find (i) the tension in the wire, and (ii) the stress in the wire.

$$\text{Volume of aluminium (V)} = \frac{\text{mass}}{\text{density}} = \frac{m}{\rho}$$
$$= \frac{11{\cdot}5 \times 10^3}{2{\cdot}56} \text{ cm}^3$$

$$\text{Mass of an equal vol. of oil} = \text{mass of oil displaced}$$
$$= V \times \rho$$
$$= \frac{11{\cdot}5 \times 10^3}{2{\cdot}56} \times 0{\cdot}9 \text{ g}$$
$$= \frac{11{\cdot}5 \times 0{\cdot}9}{2{\cdot}56} \text{ kg}$$

$$\text{Wt. of aluminium in oil} = \text{wt. in air} - \text{wt. of oil displaced}$$
$$= 11{\cdot}5 \times 9{\cdot}81 - \frac{11{\cdot}5 \times 0{\cdot}9 \times 9{\cdot}81}{2{\cdot}56}$$
$$= 11{\cdot}5 \times 9{\cdot}81 \left\{1 - \frac{0{\cdot}9}{2{\cdot}56}\right\}$$
$$= 11{\cdot}5 \times 9{\cdot}81 \left\{\frac{2{\cdot}56 - 0{\cdot}9}{2{\cdot}56}\right\}$$

$$= \frac{11{\cdot}5 \times 9{\cdot}81 \times 1{\cdot}66}{2{\cdot}56}$$

$$= 73{\cdot}16 \text{ N} \quad \text{Ans. (i)}$$

$$\text{Stress} = \frac{\text{load}}{\text{area}}$$

$$= \frac{73{\cdot}16}{0{\cdot}7854 \times 1{\cdot}5^2} = 41{\cdot}4 \text{ N/mm}^2$$

$$= 41{\cdot}4 \times 10^6 \text{ N/m}^2$$

$$= 41{\cdot}4 \text{ MN/m}^2 \quad \text{Ans. (ii)}$$

FLOATING BODIES

A body which floats freely in a liquid is in equilibrium, therefore its downward weight must be equal to the upward force of buoyancy. The upward thrust is equal to the weight of liquid displaced, hence a *floating body displaces an amount of liquid equal to its own weight.*

All cases of floating bodies can be dealt with by the simple law of equilibrium:

Total downward force = Total upward force

Example. A rectangular block of wood 375 mm long by 250 mm broad by 100 mm deep, floats in fresh water. If the density of the wood is $0{\cdot}75$ g/cm³, find the draught at which it floats.

$$\text{Let } d = \text{draught}$$

$$\text{Downward force} = \text{Upward force}$$

$$\text{Weight of wood} = \text{Weight of water displaced}$$

$$37{\cdot}5 \times 25 \times 10 \times 0{\cdot}75 \times 10^{-3} \times 9{\cdot}81 = 37{\cdot}5 \times 25 \times d \times 1 \times 10^{-3} \times 9{\cdot}81$$

$$10 \times 0{\cdot}75 = d$$

$$d = 7{\cdot}5 \text{ cm} = 75 \text{ mm} \quad \text{Ans.}$$

Example. A solid wood raft 4 m long by $1{\cdot}6$ m broad by $0{\cdot}6$ m deep, floats at a draught of $0{\cdot}5$ m in sea water when carrying a mass of 592 kg on top of the raft. Find the density of the wood, taking the density of sea water as 1025 kg/m³.

$$\text{Downward force} = \text{Upward force}$$

$$\text{Wt. of raft} + \text{wt. of mass} = \text{wt. of water displaced}$$

$$4 \times 1{\cdot}6 \times 0{\cdot}6 \times \rho \times 9{\cdot}81 + 592 \times 9{\cdot}81 = 4 \times 1{\cdot}6 \times 0{\cdot}5 \times 1{\cdot}025 \times 10^3 \times 9{\cdot}81$$

$$3{\cdot}84 \times \rho + 592 = 3280$$

$$3{\cdot}84\rho = 2688$$

$$\rho = 700 \text{ kg/m}^3$$

Density of wood = 700 kg/m^3 or 0·7 t/m^3 or 0·7 g/cm^3 Ans.

Example. The mass of a cork lifebuoy is 12·25 kg. Find the maximum mass of cast iron it can support in sea water if the iron is suspended below the buoy. Take the relative density of cork, cast iron, and sea water, as 0·288, 7·21 and 1·025 respectively.

Maximum load will be carried when the buoy is just awash.

$$\text{Volume of buoy} = \frac{\text{mass}}{\text{density}}$$

$$= \frac{12{\cdot}25 \times 10^3}{0{\cdot}288} = 42{\cdot}54 \times 10^3 \text{ cm}^3$$

Mass of sea water displaced by buoy when fully submerged

$$= \text{volume} \times \text{density of sea water}$$

$$= 42{\cdot}54 \times 10^3 \times 1{\cdot}025 \text{ g}$$

$$= 43{\cdot}6 \text{ kg}$$

$$\text{Volume of cast iron} = \frac{\text{mass}}{\text{density}}$$

$$= \frac{m \times 10^3}{7{\cdot}21} \text{ cm}^3$$

Mass of sea water displaced by cast iron

$$= \frac{m \times 10^3}{7{\cdot}21} \times 1{\cdot}025 \text{ g}$$

$$= 0{\cdot}1422\, m \text{ kg}$$

$$\text{Total downward forces} = \text{Total upward forces}$$

$$\text{Weights of buoy \& cast iron} = \text{Weight of water displaced}$$

$$9{\cdot}81\,(12{\cdot}25 + m) = 9{\cdot}81\,(43{\cdot}6 + 0{\cdot}1422m)$$

$$m - 0{\cdot}1422\, m = 43{\cdot}6 - 12{\cdot}25$$

$$0{\cdot}8578\, m = 31{\cdot}35$$

$$m = 36{\cdot}55 \text{ kg} \quad \text{Ans.}$$

PRESSURE HEAD

Consider the equilibrium of a vertical column of liquid of height h m, cross-sectional area A m^2, and of density ρ kg/m^3, as illustrated in Fig. 204.

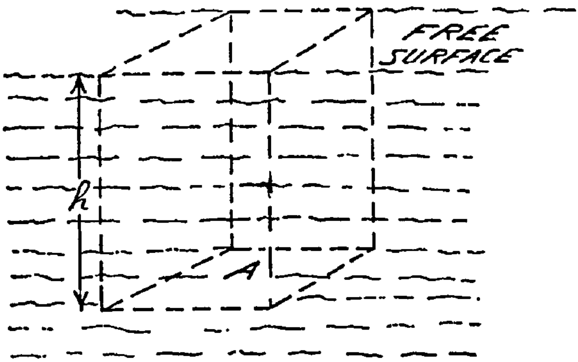

Fig. 204

$$\begin{aligned} \text{The } \textit{mass} \text{ of the liquid column} &= \text{volume} \times \text{density} \\ &= h\,A \times \rho \\ \text{The } \textit{weight} \text{ of the liquid column} &= h\,A\,\rho \times g \end{aligned}$$

Let p = pressure in the liquid at depth h

For equilibrium:

$$\begin{aligned} \text{Upward force} &= \text{Weight of liquid column} \\ p \times A &= \rho g h A \\ \therefore p &= \rho g h \end{aligned}$$

Since the density of fresh water is 10^3 kg/m^3 the head of fresh water which will exert a pressure of 1 bar (*i.e.* 10^5 N/m^2) can be calculated:

$$\begin{aligned} p &= \rho g h \\ h &= \frac{p}{\rho g} \\ h &= \frac{10^5}{10^3 \times 9{\cdot}81} \\ h &= 10{\cdot}19 \text{ m} \end{aligned}$$

i.e. a head of 10·19 m of fresh water will exert a pressure of 1 bar.

Thus, pumping against a pressure (p) can be considered as lifting the liquid to an equivalent height (h) and the work done or power exerted can be calculated by this method.

Example. An engine developing 3730 kW uses 7·25 kg of steam/kWh. If the boiler pressure is 17 bar (= 17×10^5 N/m²), calculate the output power of the feed pump.

Equivalent water head of 17 bar $= 17 \times 10{\cdot}19 = 173{\cdot}2$ m

Mass of water pumped into boiler every second

$$= \frac{7{\cdot}25 \times 3730}{3600} = 7{\cdot}511 \text{ kg}$$

Force to lift against gravity $= 7{\cdot}511 \times 9{\cdot}81 = 73{\cdot}69$ N

$$\begin{aligned} \text{Power} &= \text{work done per second} \\ &= \text{force} \times \text{height, per second} \\ &= 73{\cdot}69 \times 173{\cdot}2 \\ &= 1{\cdot}277 \times 10^4 \text{ W} = 12{\cdot}77 \text{ kW Ans.} \end{aligned}$$

This is set out as an example of 'equivalent head'. An alternative solution to this type of example, and a more direct approach, is explained in Chapter 4, thus:

$$\begin{aligned} \text{Work done} &= p \times V \\ \text{Work per sec.} &= \text{Pressure} \times \text{Volume flow} \\ \text{Density of water} &= 10^3 \text{ kg/m}^3 \\ \text{Volume flow} &= \frac{7{\cdot}25 \times 3730}{10^3 \times 3600} = 7{\cdot}511 \times 10^{-3} \text{ m}^3\text{/s} \\ \text{Power} &= 17 \times 10^5 \times 7{\cdot}511 \times 10^{-3} \\ &= 1{\cdot}276 \times 10^4 \text{ W} = 12{\cdot}76 \text{ kW} \end{aligned}$$

MANOMETERS

A manometer is a simple pressure measuring device, suitable for measuring small pressures, or small differences in pressure.

The manometer shown in Fig. 205 can be used for measuring air pressure. It consists of a U-tube containing a liquid, one end of the U-tube is connected to an air trunk or duct, the other end is open to the atmosphere.

The difference in the level of liquid in the two limbs of the U-tube registers the difference in pressure between the two ends of the limbs. (*i.e.* it registers the amount by which the pressure in the air trunk is *above* atmospheric pressure.)

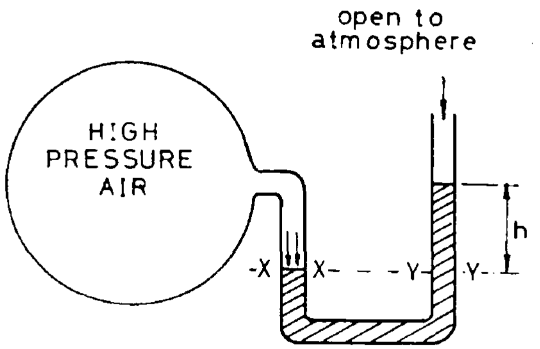

Fig. 205

For equilibrium,

$$\text{Air pressure acting at XX} = \text{pressure due to head } h \text{ of the liquid, at YY.}$$

$$\therefore \text{ Air pressure in the duct} = \rho g h$$

Example. A manometer connected to an air trunk has a head difference of 18 mm of mercury (ρ = 13·6) between the two limbs. Calculate the air pressure in the trunk.

$$\begin{aligned} \text{Air pressure} &= \rho g h \\ &= 13{\cdot}6 \times 10^3 \times 9{\cdot}81 \times 0{\cdot}018 \\ &= 2{\cdot}4 \text{ kN/m}^2 \text{ Ans.} \end{aligned}$$

When a mercury manometer is used for measuring the difference in pressure between two points in a venturi meter, it is sometimes 'submerged' as illustrated in Fig. 206, and each leg of the manometer is full of water above the mercury columns. For this particular case, referring to Fig. 206 and considering the pressure at level LL, the mercury below this level is in equilibrium, therefore the pressure in one leg at level LL is equal to the pressure in the other leg at the same level.

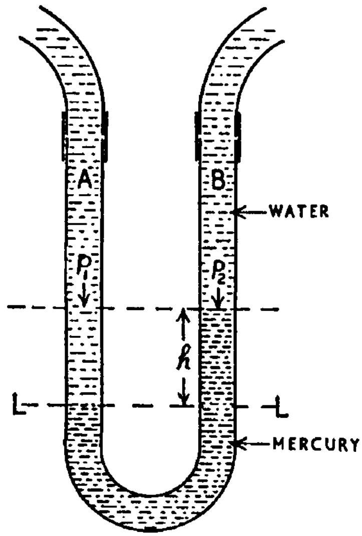

Fig. 206

Let p_1 = pressure of water in leg A

p_2 = pressure of water in leg B

h = difference in mercury level

p_1 + press. due to h metres of water = p_2 + press. due to h metres of mercury.

$p_1 - p_2$ = press. due to h metres of mercury – press. due to h metres of water

$= \rho_m g h - \rho_w g h$

$= 13{\cdot}6 \times 10^3 \times 9{\cdot}81 \times h - 1 \times 10^3 \times 9{\cdot}81 \times h$

$= 9{\cdot}81 \times 10^3 \times h \times (13{\cdot}6 - 1)$

i.e. $p_1 - p_2 = \text{h} \times 9{\cdot}81 \times 12{\cdot}6 \times 10^3$

LOAD ON AN IMMERSED SERVICE

Consider a plane of area A totally immersed in a liquid, lying at some angle θ to the surface of the liquid, as shown in Fig. 207.

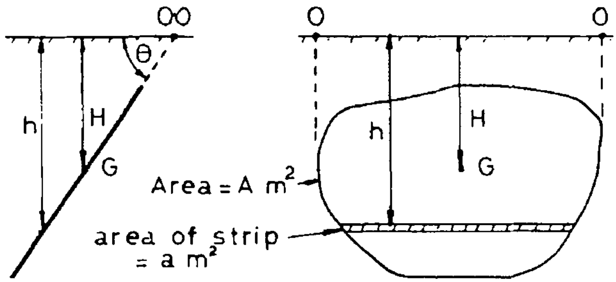

Fig. 207

The area may be divided into a large number of strips, such as the strip of area a at depth h. If the strip is very thin, the variation in pressure with depth can be ignored.

$$\text{Load on one strip} = \text{force} \times \text{area}$$

$$= \rho g h \times a$$

$$\text{Load on plane} = \rho g h_1 a_1 + \rho g h_2 a_2 + \rho g h_3 a_3 + \text{etc.}$$

$$= \rho g \sum ah$$

But, Σah is the first moment of the whole area about the surface at O.

$$i.e.\ \Sigma ah = AH$$

$$\text{Hydrostatic load} = HA\rho g$$

A immersed area, H rectangular depth to its centroid from free surface.

Example. A vertical rectangular bulkhead is 7 m wide and has fresh water to a height of 6 m on one side. Calculate the water load on the bulkhead. (Density of fresh water = 1000 kg/m³).

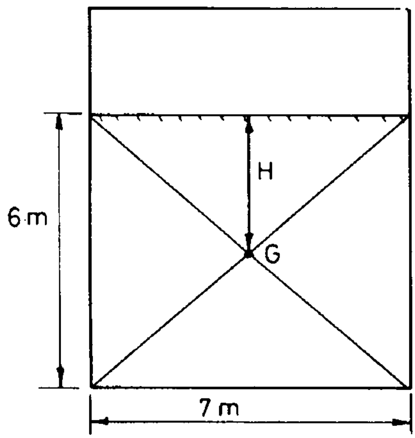

Fig. 208

$$\text{Hydrostatic load} = HA\rho g$$
$$= \frac{6}{2} \times 7 \times 6 \times 10^3 \times 9{\cdot}81$$
$$= 1{\cdot}236 \times 10^6 \text{ N}$$
$$= 1{\cdot}236 \text{ MN} \quad \text{Ans.}$$

Example. A tank 10 m long 4 m wide and 6 m high is filled with oil ($\rho = 0{\cdot}9$) and the oil rises to a height of 5 m up a vent pipe above the top of the tank. Calculate the load on one end plate and on the bottom of the tank.

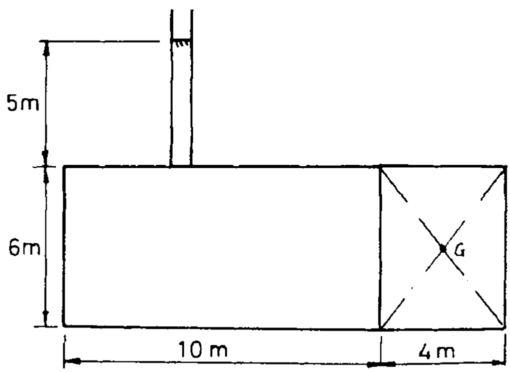

Fig. 209

$$\text{End plate : load} = HA\rho g$$
$$= 8 \times 4 \times 6 \times 900 \times 9{\cdot}81$$
$$= 1{\cdot}695 \text{ MN} \quad \text{Ans.}$$

$$\text{Bottom plate : load} = HA\rho g$$
$$= 11 \times 4 \times 10 \times 900 \times 9{\cdot}81$$
$$= 3{\cdot}885 \text{ MN} \quad \text{Ans.}$$

TRANSMISSION OF POWER

The intensity of fluid pressure at a given depth is the same in all directions and acts perpendicularly (at right angles) to the walls of its container. In closed high pressure systems such as in steering units, hydraulic water-tight door circuits, hydraulic jacks, etc., the difference in pressure due to difference in height is negligible in comparison with the high working pressure and therefore the intensity of pressure can be taken as being uniform throughout the whole system.

The hydraulic jack is a typical example to illustrate transmission of hydraulic pressure and this was dealt with in Chapter 8.

CENTRE OF PRESSURE

The centre of pressure on an immersed area is the point at which the total liquid load may be regarded as acting.

Again considering a plane of area A immersed in a liquid, at an angle θ to the surface, as in Fig. 210:

Let y = distance from the strip to the surface at O.

$$\text{It is seen that } h = y \sin\theta$$
$$\text{Load on strip} = ha\rho g$$
$$= y \sin\theta \, a\rho g$$
$$\text{Load on plane} = \rho g \sin\theta \, (a_1 y_1 + a_2 y_2 + a_3 y_3 + \text{etc.})$$
$$= \rho g \sin\theta \, \Sigma ay$$

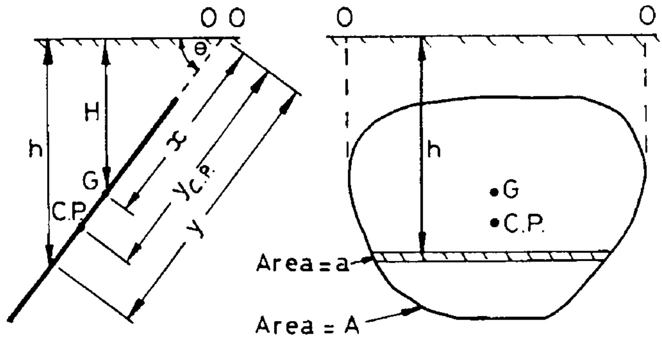

Fig. 210

Taking moments about the surface at O,

$$\textit{Moment} \text{ of load on the strip} = y \times \rho g a y \sin\theta = \rho g a y^2 \sin\theta$$

$$\textit{Moment} \text{ of load on the plane} = \rho g \sin\theta\,(a_1 y_1 + a_2 y_2 + a_3 y_3 + \text{etc.}) = \rho g \sin\theta\,\Sigma a y^2$$

$$\text{Distance to C.o.P. from } O = \frac{\text{Moment of load}}{\text{load}} = \frac{\rho g \sin\theta\,\Sigma a y^2}{\rho g \sin\theta\,\Sigma a y} = \frac{\Sigma a y^2}{\Sigma a y}$$

Now, the term Σay is the *first* moment of the area of the plane about the surface at O and Σay^2 is the *second* moment of the area about O.

$$y_{cp} = \frac{\text{second moment of the area about the surface}}{\text{first moment of the area about the surface}}$$

Where y_{cp} is the projected distance from the centre of pressure to the surface at O.

The *second moment* of area about the surface may be calculated using the theorem of parallel axes (see Chapter 7).

$$I_O = I_G + Ax^2$$

where I_O = second moment of the area about the surface at O

I_G = second moment of the area about an axis through its centroid at G

A = immersed area

x = distance between the axis through O and the axis through G

$$\text{hence, } y_{cp} = \frac{I_O}{Ax}$$

$$y_{cp} = \frac{I_G + Ax^2}{Ax}$$

$$y_{cp} = \frac{I_G}{Ax} + x$$

$$x = \frac{H}{\sin\theta}$$

$$y_{cp} = \frac{I_G}{AH/\sin\theta} + \frac{H}{\sin\theta}$$

If the immersed area is *vertical,* then angle $= 90°$, $\sin\theta = 1$

Hence, for a vertical area $y_{cp} = \dfrac{I_G}{AH} + H$

Two common shapes for tanks or bulkheads are rectangular and inverted triangular areas.

(i) Rectangular area with its upper edge in the surface of the liquid (Fig. 211):

For a rectangular area as shown,

$$I_G = \frac{BD^3}{12}$$

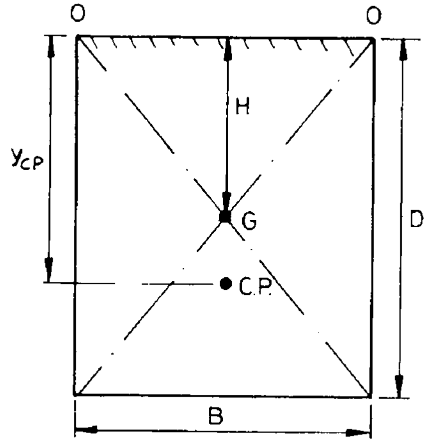

Fig. 211

$$y_{cp} = \frac{I_G}{AH} + H$$

$$= \frac{B \times D^3}{12 \times B \times D \times \frac{1}{2}D} + \frac{D}{2}$$

$$= \frac{D}{6} + \frac{D}{2}$$

$$= \tfrac{2}{3}D$$

Centre of pressure is at ⅔ depth below the surface of the liquid.

(ii) Inverted triangular area with its upper edge in the surface of the liquid (Fig. 212).

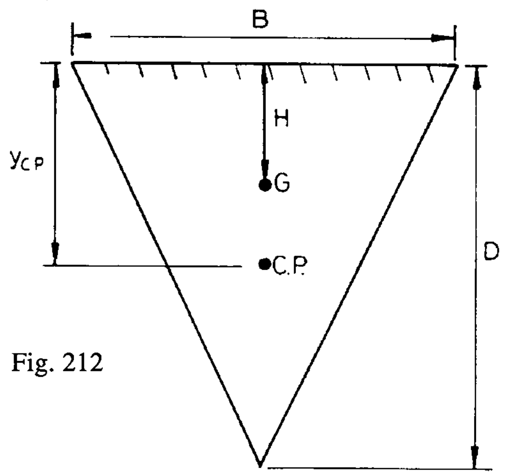

Fig. 212

For a triangular area as shown,

$$I_G = \frac{BD^3}{36}$$

(all such I values are determined by integration).

$$y_{cp} = \frac{I_G}{AH} + H$$

$$= \frac{B \times D^3}{36 \times \frac{1}{2} \times B \times D}$$

$$= \frac{D}{6} + \frac{D}{3}$$

$$= \tfrac{1}{2}D$$

Centre of pressure is at ½ depth below the surface of the liquid.

Example. A tank with sides 6 m long contains fresh water (ρ = 1000 kg/m^3) to a depth of 7 m. Calculate (a) the force on one side of the tank and (b) determine the position of the centre of pressure.

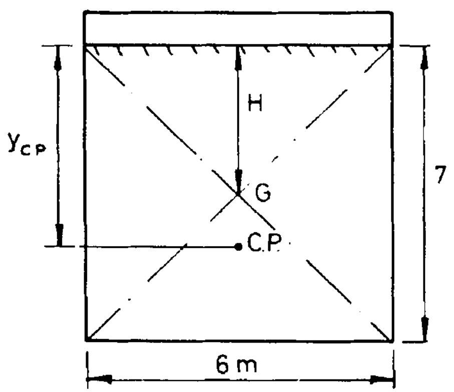

Fig. 213

(a) Force $= HA\rho g$

$= 3{\cdot}5 \times 7 \times 6 \times 10^3 \times 9{\cdot}81$

$= 1{\cdot}442$ MN Ans.

(b) y_{cp} = ⅔ depth below the surface

Position of C.o.P = 4⅔ m below the liquid surface. Ans.

Example. A rectangular plate, 3 m wide and 2 m high is fitted in the vertical side of a tank. The tank contains fresh water to a height of 7 m above the top edge of the plate. Calculate (a) the load on the plate and (b) the position of the centre of pressure.

(Density of F.W. = 1000 kg/m³)

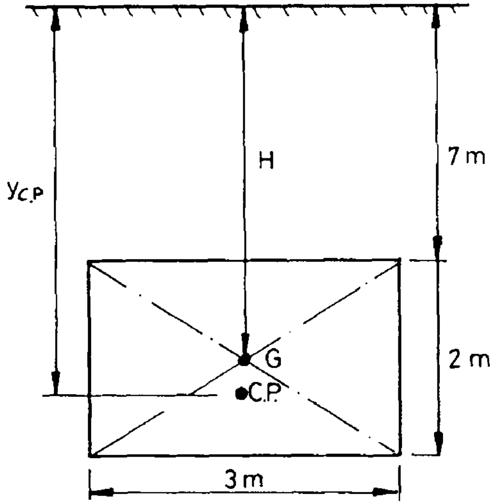

Fig. 214

(a) Load on plate $= HA\rho g$

$= 8 \times 2 \times 3 \times 1000 \times 9{\cdot}81$

$= 470{\cdot}9$ kN Ans.

(b) Note: in the problem, the upper edge of the rectangular area is *not* in the free surface of the liquid.

$$y_{cp} = \frac{I_G}{AH} + H$$

$$= \frac{3 \times 2^3}{12 \times 3 \times 2 \times 8} + 8$$

$$= 8{\cdot}0417$$

Position of C.o.P = 8·0417 m below the surface

f HYDROSTATIC FORCE AND CENTRE OF PRESSURE FOR NON-MIXING LIQUIDS

If a tank contains two non-mixing liquids the resultant force and position of centre of pressure may be determined as shown below.

Example. A tank with sides 2 m long contains 1 m of sea water ($\rho = 1{\cdot}024$) with 0·6 m of oil ($\rho = 0{\cdot}8$) above.

Calculate the force on one side of the tank and the position of the centre of pressure.

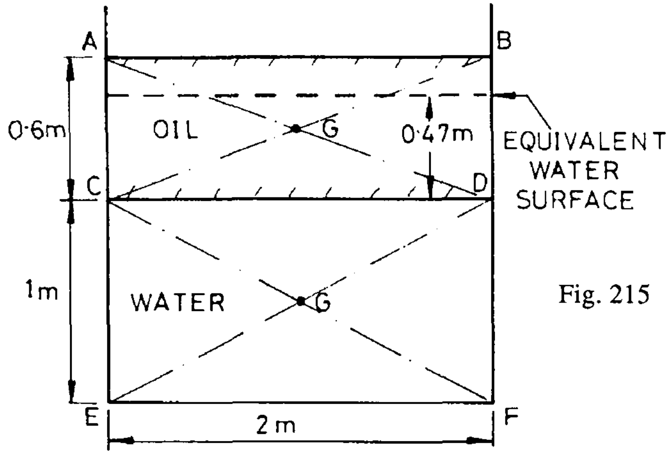

Fig. 215

(a) Area ABCD:

$$\begin{aligned} \text{oil load} &= HA\rho g \\ &= 0{\cdot}3 \times 2 \times 0{\cdot}6 \times 800 \times 9{\cdot}81 \\ &= 2{\cdot}825 \text{ kN} \\ y_{cp} &= \tfrac{2}{3} \text{ depth below the oil surface} \\ &= 0{\cdot}4 \text{ m below the oil surface.} \end{aligned}$$

(b) Area CDEF:

Note: Sea water is acting on the area CDEF and the superimposed liquid (oil) can be converted to an *equivalent depth* of sea water.

For the oil, equivalent depth of S.W. $= 0{\cdot}6 \times \dfrac{800}{1024}$

$= 0{\cdot}47$ m

Thus, an equivalent water surface at 0·47 m above CD is used in calculating the water load on CDEF.

$$\text{Water load} = HA\rho g$$
$$= (0{\cdot}5 + 0{\cdot}47) \times 1 \times 2 \times 1024 \times 9{\cdot}81$$
$$= 19{\cdot}49 \text{ kN}$$

$$y_{cp} = \frac{I_G}{AH} + H$$
$$= \frac{2 \times 1^3}{12 \times 1 \times 2 \times 0{\cdot}97} + 0{\cdot}97$$
$y_{cp} = 1{\cdot}056$ m below the equivalent water surface.

Position of C.o.P $= 1{\cdot}056 + (0{\cdot}6 - 0{\cdot}47)$

$= 1{\cdot}186$ m below the surface at AB.

(c) Thus, for the total area ABFE,

The total load $= 2{\cdot}825 + 19{\cdot}49$

$= 22{\cdot}315$ kN. Ans.

Let y_{cp} = depth to the resultant C.o.P from the surface at AB

Taking moments about the surface:

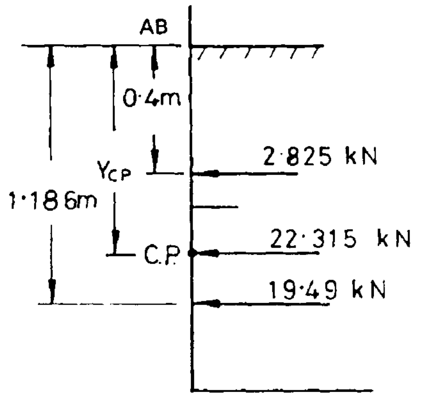

Fig. 216

$$22{\cdot}315 \times y_{cp} = 2{\cdot}825 \times 0{\cdot}4 + 19{\cdot}49 \times 1{\cdot}186$$
$$y_{cp} = 1{\cdot}086 \text{ m}$$

Position of resultant C.o.P = 1·086 m below the surface. Ans.

TEST EXAMPLES 12

1. Three liquids of specific gravities 0·7, 0·8 and 0·9 respectively are mixed in the proportion of 3, 4 and 5 by mass. Find the relative density of the mixture.

2. A piece of brass is suspended by a wire from a spring balance and it registers a weight of 7 N in air. When the brass is lowered into fresh water until submerged the balance shows a weight of 6·17 N. Find the relative density of this brass.

3. A plank of wood of relative density 0·6 is 3 m long by 300 mm broad by 150 mm thick and floats in fresh water at a draught of 125 mm when a casting is carried on top of the plank. Calculate the mass of the casting.

4. A solid wood raft 7·6 m long by 1·22 m wide by 230 mm thick, of relative density 0·75, floats in sea water. Find the draught when a piece of lead of mass 508 kg and relative density 11·4 is suspended beneath the raft. Take density of sea water as 1025 kg/m^3.

5. Find the relative density of an oil if a vertical column of this oil 1·511 m high will balance a vertical column of mercury 10 cm high. Take the relative density of mercury as 13·6.

6. A vertical bulkhead 9·6 m wide and 6 m deep separates two tanks. One tank contains oil of relative density 0·85 to a depth of 5·4 m, and the other tank is empty. Find the total load on the bulkhead in MN and state the position of the centre of pressure.

7. Two tanks are separated by a vertical rectangular bulkhead 7·5 m wide. In one tank there is fresh water to a depth of 7·2 m and in the other there is oil of relative density 0·8 to a depth of 4·2 m. Find the resultant thrust on the bulkhead in MN and the position of the resultant centre of pressure.

8. A rectangular tank is 2 m long, 1·2 m wide and 1m deep, and contains fresh water to a depth of 0·6 m. A block of wood of mass 156 kg is put into the water and floats freely. Find the water load on the bottom, sides and ends of the tank, in KN.

*f*9. Prove that hydrostatic force = $HA\rho g$ on an immersed area A, inclined at any angle below the free surface of a liquid.

*f*10. A thin hollow metal sphere, diameter 2 m, is filled with water (density 1000 kg/m^3). Determine (a) maximum pressure (b) total load (c) resultant force on the interior surface of the sphere.

*f*11. Derive the value of the second moment of area through the centroid about a diameter (I_G) for a circular plate and hence determine the position of the centre of pressure for such a plate with its edge in the surface of a liquid (*xx*).

*f*12. A circular door, 1·5 m diameter, lies with its plane inclined at 35° to the horizontal and its upper edge is 1 m below the surface of the water (density 1000 kg/m^3). Calculate the hydrostatic load on the door and the position of the centre of pressure.

*f*13. A bulkhead is in the form of a triangle, 6 m wide at the top and 9 m deep. the tank is filled with sea water (density 1025 kg/m^3). Calculate the load on the bulkhead and the position of the centre of pressure relative to the top of the bulkhead if water is: (a) at top of the bulkhead (b) 4 m up a sounding pipe.

*f*14. A plate covers a rectangular hole 30 cm wide by 60 cm high in a vertical side of a tank which contains water (density 1000 kg/m^3) which is level with the top edge of the hole. Determine the load on the plate and the position of the centre of pressure relative to the top edge of the hole. If the water level is raised by 1 m what will be the new position of the centre of pressure?

*f*15. A closed cylindrical tank is 1 m diameter and 2 m high. It contains water (density 1000 kg/m^3) to a depth of 1·5 m and the space above the water contains air at a pressure of 120 kN/m^3.

(a) Calculate the resultant force on the vertical seam of the tank.

(b) Determine the vertical height to the centre of pressure from the bottom of the tank.